

Windows Command Line Basics

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These target IP addresses were meant to be pinged in the lab. I couldn't reach them so decided to substitute them in with 8.8.8.8 and 1.1.1.1 which are Google and CloudFlare respectively.

Target Linux Systems

- 172.15.1.113
- 172.15.1.101
- 172.15.1.104

Target Windows XP Systems

- 172.15.1.105
- 172.15.1.100
- 172.15.1.107

1. Checking IP Configuration (ipconfig)

```
C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::6404:4036:be8f:fd15%10
    IPv4 Address. . . . . : 172.16.200.106
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.200.254
```

2. Displaying Detailed Network Information (ipconfig /all)

```

C:\Windows\System32>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-NP36IIS
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . . :
    Description . . . . . : Intel(R) PRO/1000 MT Network Connection
    Physical Address. . . . . : BC-24-11-0C-6A-10
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::6404:4036:be8f:fd15%10(Preferred)
    IPv4 Address. . . . . : 172.16.200.106(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Wednesday, November 12, 2025 4:14:16 PM
    Lease Expires . . . . . : Wednesday, November 19, 2025 4:14:16 PM
    Default Gateway . . . . . : 172.16.200.254
    DHCP Server . . . . . : 172.16.200.254
    DHCPv6 IAID . . . . . : 112993297
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-54-DB-C7-BC-24-11-0C-6A-10
    DNS Servers . . . . . : 96.45.45.45
                           96.45.46.46
    NetBIOS over Tcpip. . . . . : Enabled

```

3. Releasing the IP Address (ipconfig /release)

```

C:\Windows\System32>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . . :
    Link-local IPv6 Address . . . . . : fe80::6404:4036:be8f:fd15%10
    Default Gateway . . . . . :

```

4. Renewing the IP Address (ipconfig /renew)

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::6404:4036:be8f:fd15%10
    IPv4 Address. . . . . : 172.16.200.106
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.200.254
```

5. Flushing the DNS Cache (ipconfig /flushdns)

```
C:\Windows\System32>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```

6. Checking Host Connectivity (ping)

```
C:\Windows\System32>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=26ms TTL=116
Reply from 8.8.8.8: bytes=32 time=20ms TTL=116
Reply from 8.8.8.8: bytes=32 time=24ms TTL=116
Reply from 8.8.8.8: bytes=32 time=21ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 26ms, Average = 22ms
```

```
C:\Windows\System32>ping 1.1.1.1

Pinging 1.1.1.1 with 32 bytes of data:
Reply from 1.1.1.1: bytes=32 time=18ms TTL=54
Reply from 1.1.1.1: bytes=32 time=17ms TTL=54
Reply from 1.1.1.1: bytes=32 time=20ms TTL=54
Reply from 1.1.1.1: bytes=32 time=21ms TTL=54

Ping statistics for 1.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 21ms, Average = 19ms
```

7. Checking Network Path (pathping)

```
C:\Windows\System32>pathping 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:
 0  DESKTOP-NP36IIS [172.16.200.106]
 1  172.16.200.254
 2  50.213.68.70
 3  96.120.74.233
 4  ae-252-1209-rur01.mtlaurel.nj.panjde.comcast.net [24.124.225.193]
 5  ae-2-rur02.mtlaurel.nj.panjde.comcast.net [68.85.63.18]
 6  be-14-rar01.absecon.nj.panjde.comcast.net [68.85.62.181]
 7  be-101-ar03.plainfield.nj.panjde.comcast.net [96.108.49.105]
 8  * * *
Computing statistics for 175 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
     Lost/Sent = Pct Lost/Sent = Pct  Lost/Sent = Pct
 0      0ms      0/ 100 = 0%      0/ 100 = 0%      DESKTOP-NP36IIS [172.16.200.106]
 1      0ms      0/ 100 = 0%      0/ 100 = 0%      172.16.200.254
 2      0ms      0/ 100 = 0%      0/ 100 = 0%      50.213.68.70
 3     10ms      0/ 100 = 0%      0/ 100 = 0%      96.120.74.233
 4     15ms      0/ 100 = 0%      0/ 100 = 0%      ae-252-1209-rur01.mtlaurel.nj.panjde.comcast.net [24.124.225.193]
 5     15ms      0/ 100 = 0%      0/ 100 = 0%      ae-2-rur02.mtlaurel.nj.panjde.comcast.net [68.85.63.18]
 6     15ms      0/ 100 = 0%      0/ 100 = 0%      be-14-rar01.absecon.nj.panjde.comcast.net [68.85.62.181]
 7     23ms      0/ 100 = 0%      0/ 100 = 0%      be-101-ar03.plainfield.nj.panjde.comcast.net [96.108.49.105]

Trace complete.
```

```
Trace complete.

C:\Windows\System32>pathping 1.1.1.1

Tracing route to one.one.one.one [1.1.1.1]
over a maximum of 30 hops:
 0  DESKTOP-NP36IIS [172.16.200.106]
 1  172.16.200.254
 2  50.213.68.70
 3  96.120.74.233
 4  ae-252-1209-rur01.mtlaurel.nj.panjde.comcast.net [24.124.225.193]
 5  ae-2-rur02.mtlaurel.nj.panjde.comcast.net [68.85.63.18]
 6  be-14-rar01.absecon.nj.panjde.comcast.net [68.85.62.181]
 7  be-101-ar03.plainfield.nj.panjde.comcast.net [96.108.49.105]
 8  be-31143-cs04.newark.nj.ibone.comcast.net [96.110.42.45]
 9  be-2411-pe11.newark.nj.ibone.comcast.net [96.110.33.254]
10  * * *
```

8. Tracing Route to a Destination (tracert)

```
C:\Windows\System32>tracert 1.1.1.1

Tracing route to one.one.one.one [1.1.1.1]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    172.16.200.254
  1  1 ms      1 ms      <1 ms    50.213.68.70
  2  10 ms     9 ms      11 ms    96.120.74.233
  3  27 ms     8 ms      88 ms    ae-252-1209-rur01.mtlaurel.nj.panjde.comcast.net [24.124.225.193]
  4  9 ms      13 ms     16 ms    ae-2-rur02.mtlaurel.nj.panjde.comcast.net [68.85.63.18]
  5  12 ms     18 ms     12 ms    be-14-rar01.absecon.nj.panjde.comcast.net [68.85.62.181]
  6  18 ms     17 ms     20 ms    be-101-ar03.plainfield.nj.panjde.comcast.net [96.108.49.105]
  7  18 ms     18 ms     19 ms    be-31143-cs04.newark.nj.ibone.comcast.net [96.110.42.45]
  8  18 ms     16 ms     16 ms    be-2411-pe11.newark.nj.ibone.comcast.net [96.110.33.254]
  9  *         *         *        Request timed out.
 10  16 ms     16 ms     21 ms    162.158.61.125
 11  18 ms     16 ms     16 ms    one.one.one.one [1.1.1.1]

Trace complete.
```

```
C:\Windows\System32>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    172.16.200.254
  1  1 ms      <1 ms     <1 ms    50.213.68.70
  2  11 ms     8 ms      15 ms    96.120.74.233
  3  9 ms      8 ms      9 ms     ae-252-1209-rur01.mtlaurel.nj.panjde.comcast.net [24.124.225.193]
  4  8 ms      8 ms      8 ms     ae-2-rur02.mtlaurel.nj.panjde.comcast.net [68.85.63.18]
  5  15 ms     12 ms     12 ms    be-14-rar01.absecon.nj.panjde.comcast.net [68.85.62.181]
  6  20 ms     16 ms     18 ms    be-101-ar03.plainfield.nj.panjde.comcast.net [96.108.49.105]
  7  *         *         *        Request timed out.
  8  17 ms     19 ms     17 ms    142.251.247.53
  9  21 ms     17 ms     22 ms    142.251.60.225
 10  16 ms     19 ms     16 ms    dns.google [8.8.8.8]

Trace complete.
```

9. Viewing the Routing Table (route print)

```
C:\Windows\System32>route print 1.1.1.1
=====
Interface List
10...bc 24 11 0c 6a 10 .....Intel(R) PRO/1000 MT Network Connection
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
    None
Persistent Routes:
    None

IPv6 Route Table
=====
Active Routes:
    None
Persistent Routes:
    None
```

```
C:\Windows\System32>route print 8.8.8.8
=====
Interface List
10...bc 24 11 0c 6a 10 .....Intel(R) PRO/1000 MT Network Connection
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
    None
Persistent Routes:
    None

IPv6 Route Table
=====
Active Routes:
    None
Persistent Routes:
    None
```